CORAL: Learning Amyloid Fibril Ligand Docking with Cooperative Binding Rewards

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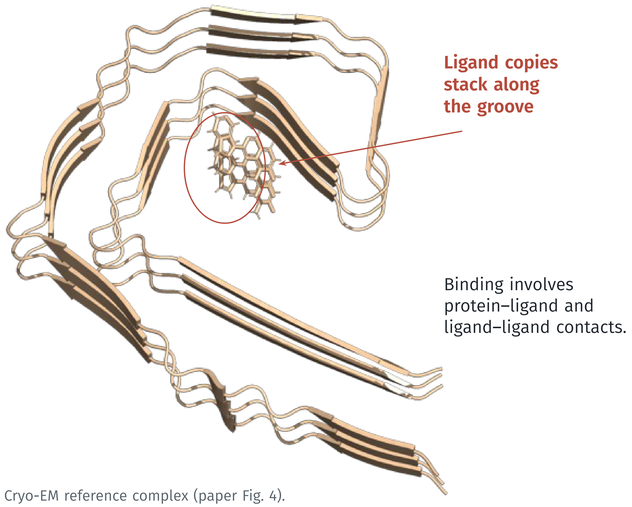
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Paper & details

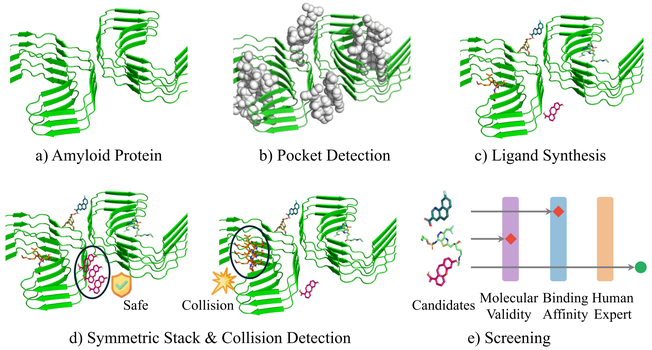
Why amyloid docking?

Amyloid fibrils form repeating cross-β grooves. The axial binding arrangement studied here contains multiple ligand copies along the fibril.



Cryo-EM reference complex (paper Fig. 4).

Curated data

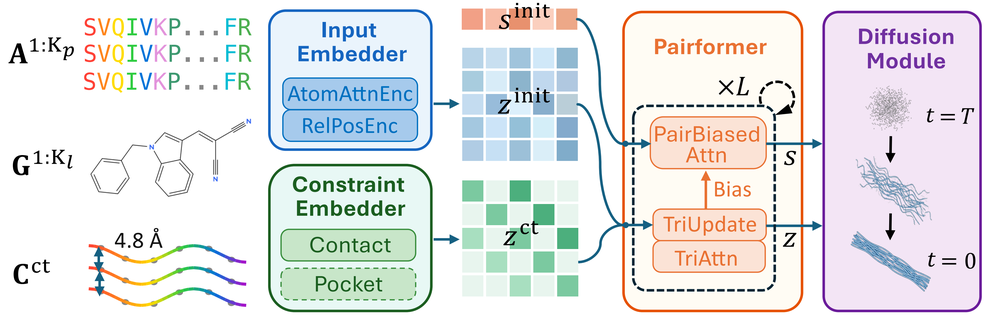


≈2,000 poses geometric warm-up training
18 complexes CryoEM evaluation
57 pairs Synthetic evaluation

Synthetic references are computational poses curated with expert review.

CORAL framework

Jointly generate all peptide chains and ligand copies, conditioned on fibril periodicity and the binding pocket.



1 Geometric warm-up

Learn periodic docking from ≈2,000 curated stacking poses.

$$\mathcal{L}_{\text{SFT}} = \alpha_{\text{diff}}\,\mathcal{L}_{\text{diff}} + \alpha_{\text{dist}}\,\mathcal{L}_{\text{dist}} + \alpha_{\text{conf}}\,\mathcal{L}_{\text{conf}}$$

Coordinate denoising, pairwise distance prediction and confidence estimation.

2 Cooperative reinforcement learning

Refine binding and ligand stacking with DiffusionNFT and a symmetry reward.

$$\mathcal{L}_{\text{RL}} = \mathcal{L}_{\text{NFT}} + \beta_{\text{KL}}\,\mathbb{E}\|v_\theta - v_{\text{ref}}\|_2^2,$$
$$\mathcal{L}_{\text{NFT}} = \mathbb{E}\big[r\,\|v_\theta^+ - \mathbf{v}\|_2^2 + (1-r)\,\|v_\theta^- - \mathbf{v}\|_2^2\big].$$

$v_\theta^\pm = (1 \mp \beta)\,v_{\text{old}} \pm \beta\,v_\theta$; v_{old} is the sampling policy.

$r \in [0,1]$: group-normalised reward; $\mathbf{v}$: target velocity.
v_{ref}: frozen warm-up model, used as a KL-style velocity anchor.

Ablation: both stages and stacking matter

AmyDock-Synthetic; medians, lower is better. Energy is a computed score.

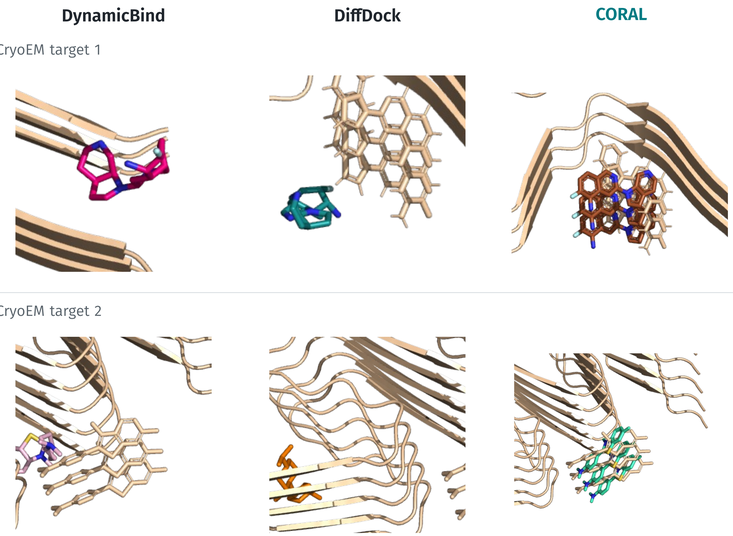
Variant	RMSD (Å)	ΔG_{eff} (kcal/mol)
CORAL (full)	**3.84**	−2.44
No stacking term	4.46	−0.95
No symmetry term	4.31	−2.21
No stability regulariser	8.13	−1.45
Warm-up only (no RL)	5.18	−1.58
RL only (no warm-up)	9.27	−0.88

Docking results

Method	CryoEM (n = 18)		Synthetic (n = 57)	
	Å↓	Hit@5↑	Å↓	Hit@5↑
DiffDock	13.68	0.0%	19.68	3.5%
DynamicBind	29.02	0.0%	17.93	1.8%
FABind	24.90	0.0%	26.87	0.0%
NeuralPLexer	19.00	5.6%	16.56	0.0%
FlowDock	21.52	0.0%	19.88	0.0%
Boltz-2	28.67	0.0%	33.87	0.0%
Chai-1	19.92	0.0%	16.52	0.0%
CORAL	**4.89**	**50.0%**	**3.84**	**68.4%**

Median heavy-atom RMSD (Å); Hit@5: RMSD ≤ 5 Å. Best predicted pose per complex, following the paper protocol.

Representative poses



Tan: reference; coloured: predictions. Original Fig. 4 close-ups; views and display scales vary.

Scope: axial stacking in two-protofilament fibrils. Remaining errors include ligand bending and orientation mismatch.

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